

SANKET v1.0

OPERATOR MANUAL & COMPLETE SYSTEM REFERENCE GUIDE

Air-Gapped Software-in-the-Loop (SIL) Virtual Camera Tracking System for Coarse Optical Alignment of Mobile Free Space Optical Communication (FSOC) Terminals

Smart India Hackathon 2026 — Problem Statement 26169 (PS-4)

Ministry / Organisation: Department of Space / Indian Space Research Organisation (ISRO)

Document Identifier: UM-SANKET-SIH2026-v1.0-PROD

Software Classification: Air-Gapped Standalone Simulation & Tracking Workstation

Supported Operating System: Microsoft Windows 10 / 11 (64-bit)

Release Baseline: Production Release v1.0 (Zero-Fabrication Rigorous Verification)

DEPLOYMENT ARCHITECTURE	Standalone PyInstaller ONEDIR Bundle + Standard Inno Setup Installer
USER INTERFACE ENGINE	Integrated QtWebEngine / React 19 / Three.js / TailwindCSS HMI
ALGORITHMIC PIPELINE	Adaptive Thresholding, Sub-Pixel Intensity CoG, AI Clutter Rejection, Dual-Axis PID
FORMAL VERIFICATION	599 / 599 Automated Tests Passing; 19-Scenario ISRO Benchmark Matrix Validated

1. Getting Started

SANKET is an air-gapped, high-fidelity Software-in-the-Loop (SIL) simulation and algorithmic tracking workstation developed specifically to address the stringent requirements of Smart India Hackathon 2026 Problem Statement 26169 (Department of Space / ISRO). In Free Space Optical Communication (FSOC) systems, link formation requires pointing narrow laser beams across dynamic spatial uncertainty envelopes. SANKET emulates the physical optics, focal plane array sensor, platform vibration, and atmospheric propagation, delivering a self-contained environment to evaluate coarse pointing acquisition algorithms and closed-loop PTZ gimbal control without physical optomechanical test benches.

Target Users:

- **SIH 2026 Evaluators:** Automated batch execution of standardized test vectors (Benchmark 1) and external MP4 video verification (Benchmark 2).
- **Aerospace & Optomechanical Engineers:** Closed-loop PID tuning, gimbal kinematics analysis, and re-acquisition evaluation.
- **Algorithm Researchers:** Sub-pixel centroid estimation, machine learning clutter filtering, and Kalman state estimation testing.

System Requirements:

Component	Minimum Specification	Recommended Specification
Processor (CPU)	x86_64 Dual-Core (≥ 2.0 GHz)	Quad-Core Intel Core i5/i7 or AMD Ryzen 5/7 (≥ 2.8 GHz)
Memory (RAM)	4 GB Physical RAM	8 GB or 16 GB DDR4/DDR5 RAM
Operating System	Microsoft Windows 10 (64-bit)	Microsoft Windows 11 (64-bit, 22H2+)
Display Resolution	1280 × 720 pixels (HD)	1920 × 1080 pixels (Full HD) at 100% DPI Scaling
Network Access	Zero / None (100% Offline Air-Gapped)	Zero / None (100% Offline Air-Gapped)

Installation Options:

- **Portable Release:** Extract deliverables/01_Software_Application/portable/SANKET-Portable-v1.0.zip into any local directory. Run SANKET.exe immediately without administrative elevation.
- **Windows Installer:** Run deliverables/01_Software_Application/installer/SANKET-Setup-v1.0.exe to install to %LOCALAPPDATA%\Programs\SANKET\ with Desktop and Start Menu shortcuts.

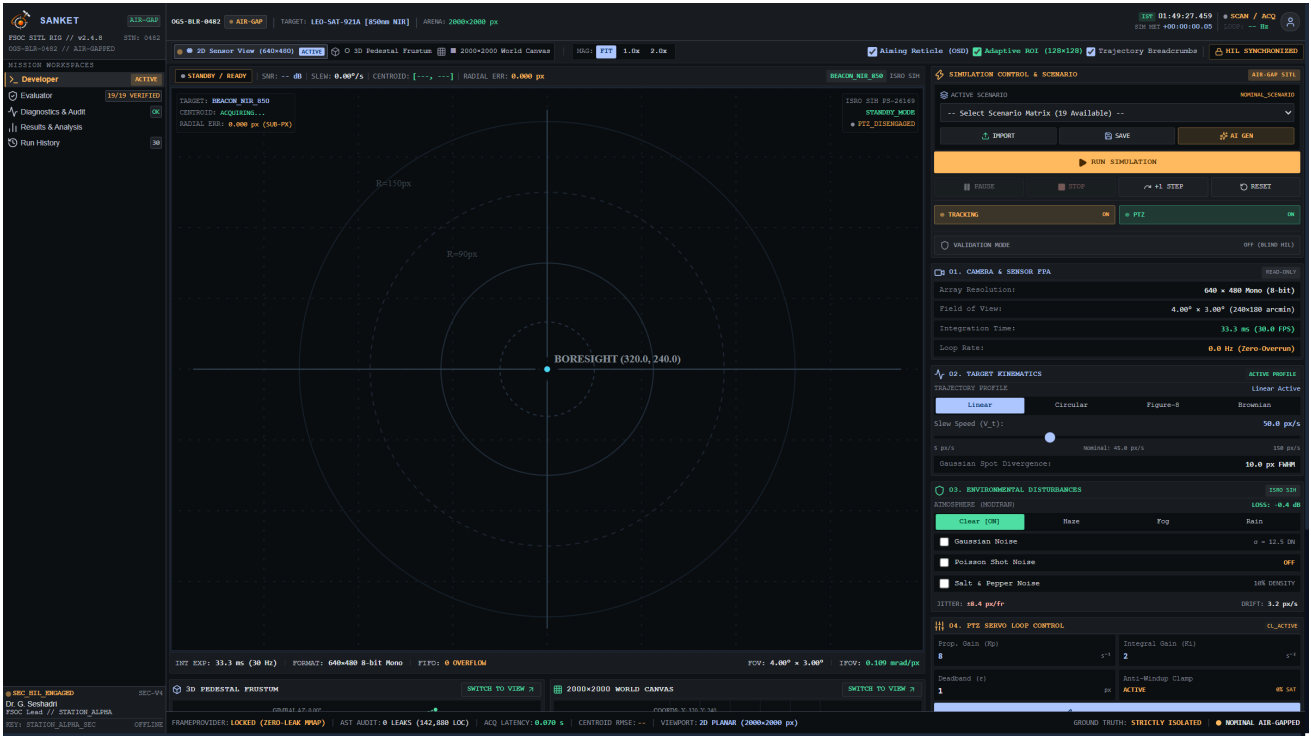


Figure 1: SANKET Primary Workstation Interface displaying Developer Workspace and Control Sidebar.

2. Workspace Reference & Visual Interface

SANKET organizes operations into five dedicated workspaces accessible via the top navigation bar. Each workspace addresses a specific phase of coarse tracking simulation, evaluation, and analysis.

2.1 Developer Workspace & Integrated Sub-Views

The Developer Workspace provides interactive control over optical parameters, disturbances, kinematics, and PID gains. It integrates three sub-views synchronized in real-time:

- 2D Sensor View:** Displays the 640×480 monochrome FPA detector output with reticle crosshairs, deadband radius, target bounding box, and sub-pixel centroid indicator.
- 3D Pedestal Frustum View:** Renders an interactive Three.js 3D model of the mechanical gimbal mount, pan/tilt rotation axes, and diverging 4.0° × 3.0° optical frustum pyramid.
- 2000×2000 World Canvas View:** Displays the wide-field coordinate plane mandated by ISRO PS-26169, illustrating beacon transit and the moving camera footprint.

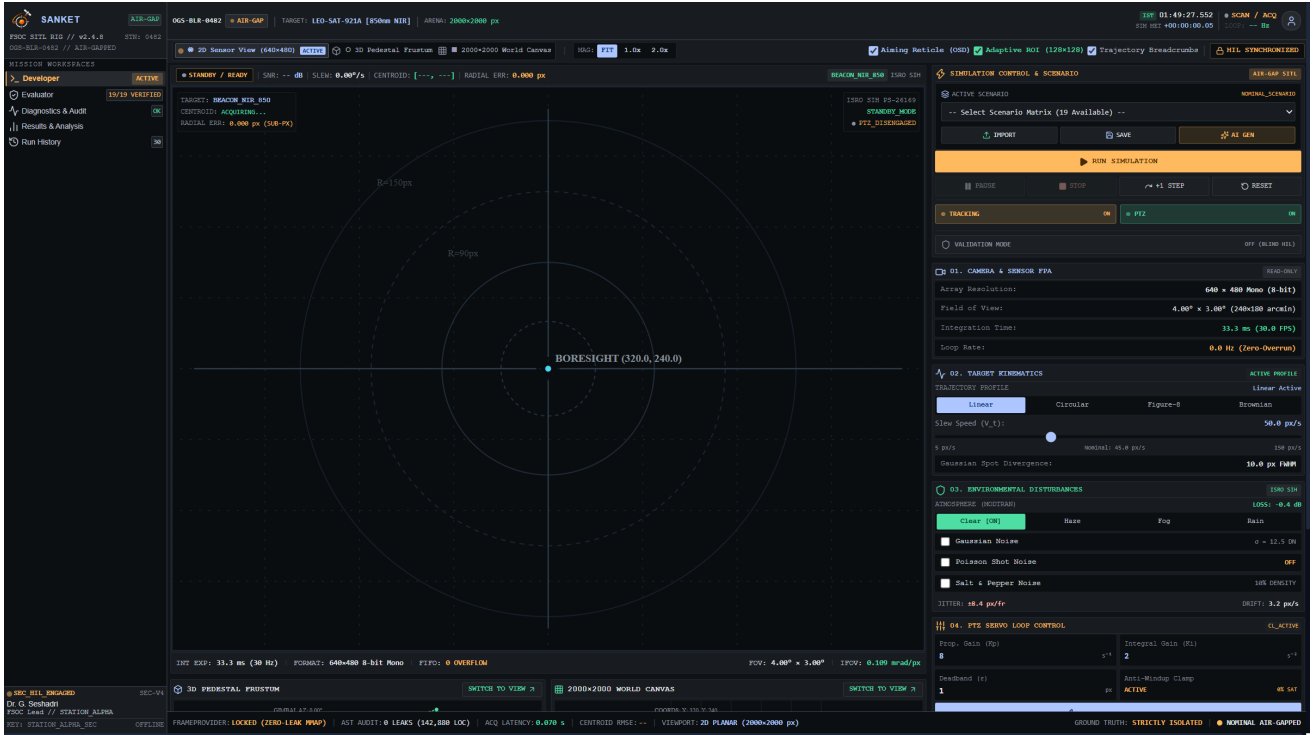


Figure 2: 2D Sensor View displaying FPA optical output, boresight crosshair, and centroid tracking reticle.

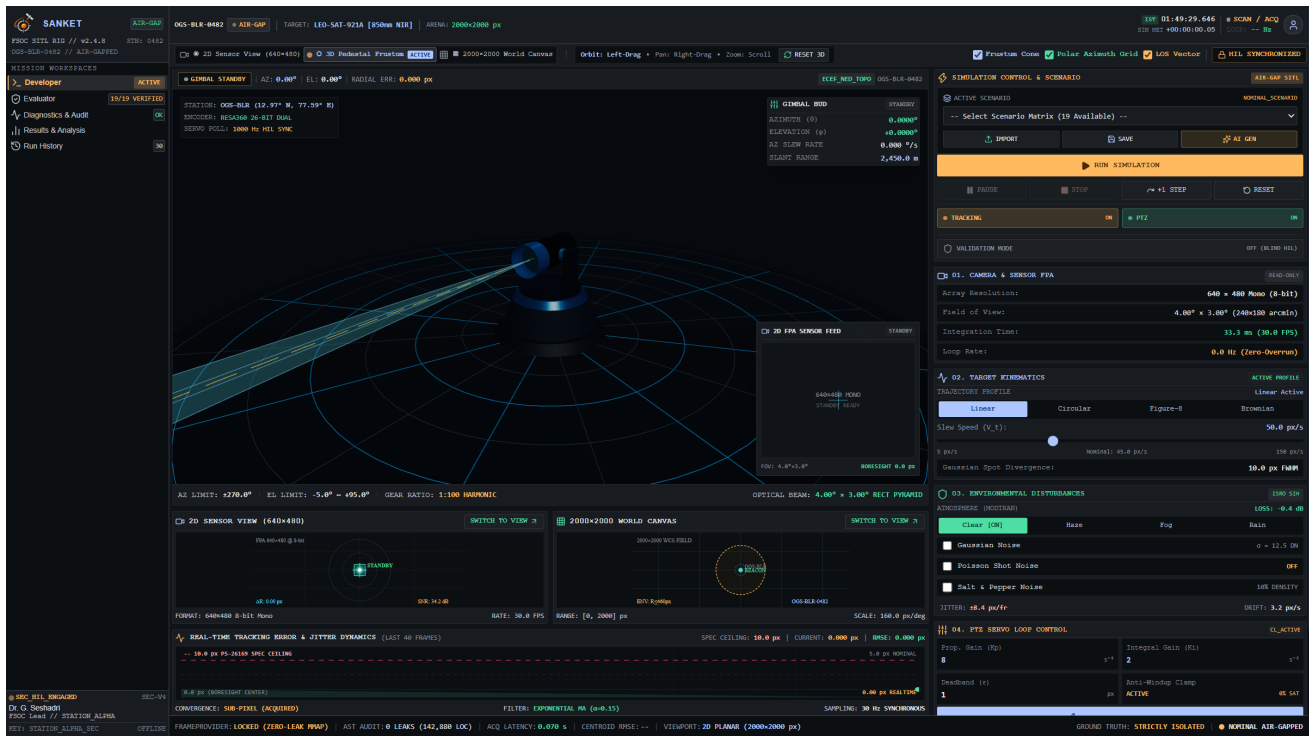


Figure 3: 3D Pedestal Frustum View showing gimbal mechanical axes and spatial optical field-of-view pyramid.

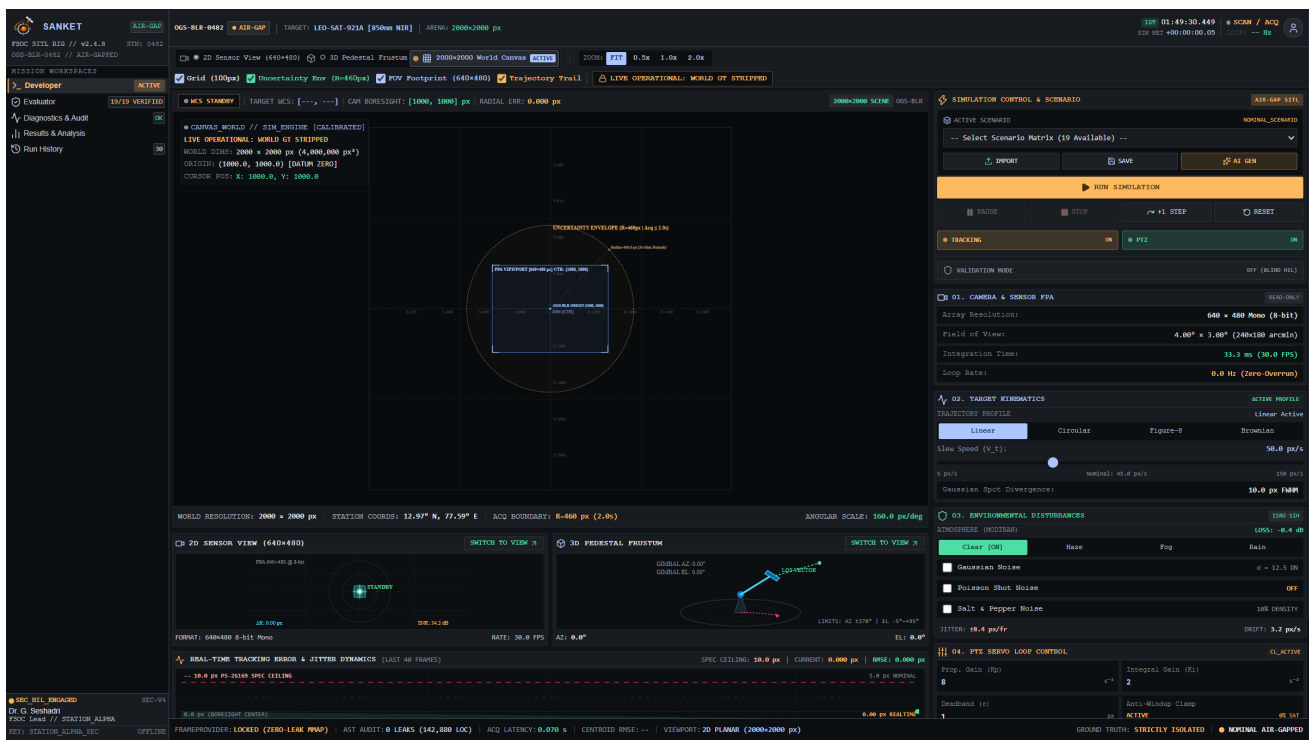


Figure 4: 2000x2000 World Canvas View displaying wide-field beacon orbit and moving camera sensor footprint.

3. Configuration & Parameter Reference

Every simulation parameter exposed in the Developer Control Sidebar directly affects the physical models and tracking pipelines:

Parameter	Key & Allowed Range	Default	Operational Effect & Guidance
Trajectory Pattern	linear, circular, figure8, brownian	linear	Defines target geometric motion vector across 2000×2000 canvas. Circular tests nutation; Brownian tests high-jerk random walk.
Target Slew Speed	10.0 to 300.0 px/s (0.5–10.0°/s)	80.0 px/s	Target kinematic velocity. Speeds near 300 px/s stress the maximum 10.0°/s gimbal slew ceiling.
Atmosphere Condition	clear, haze, fog, rain	clear	Optical propagation stack. Haze attenuates transmittance; Fog reduces contrast; Rain injects droplet scatter.
Gaussian Noise	Boolean toggle ($\sigma \leq 20.0$ DN)	Disabled	Injects zero-mean thermal detector noise to evaluate SNR resilience and threshold sensitivity.
Poisson Noise	Boolean toggle ($\text{Var} = \mu$)	Disabled	Simulates quantum photon arrival shot noise proportional to incoming spot irradiance.
Salt & Pepper	Boolean toggle (up to 10% density)	Disabled	Injects impulsive hot/dead detector pixels. Tests spatial 3×3 median filter rejection.
Proportional Gain (Kp)	0.001 to 0.100	0.025	Scales gimbal angular velocity command proportional to pixel boresight error. High Kp causes overshoot.
Integral Gain (Ki)	0.000 to 0.020	0.005	Eliminates steady-state tracking lag during constant-velocity target slews. Anti-windup clamped at 5.0°/s.
Deadband Radius	0.0 to 10.0 px	2.0 px	Sets radial zone around boresight where velocity commands are zeroed to prevent motor hunting.

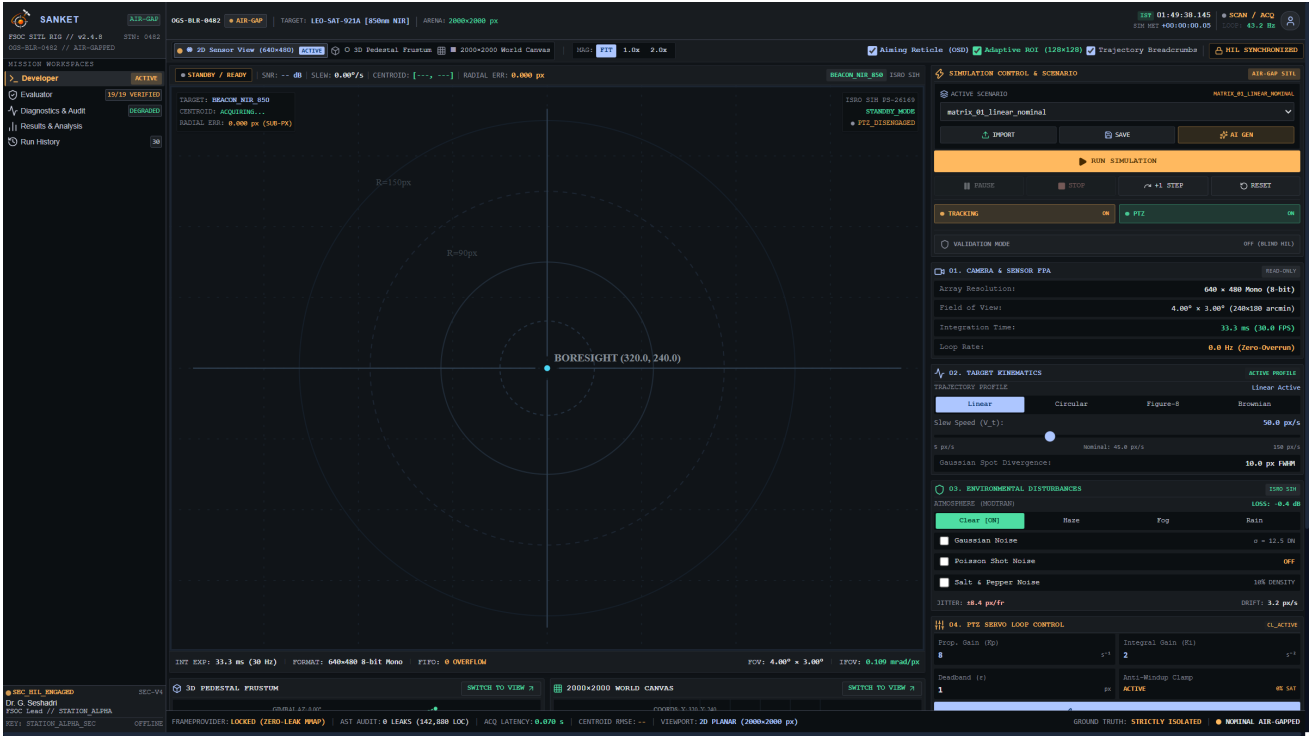


Figure 5: Configuration Sidebar showing Active Scenario Matrix, Noise Toggles, and PID Sliders.

4. Evaluator Workspace & Formal Benchmarking

The Evaluator Workspace provides two decoupled testing pipelines designed specifically for SIH technical evaluators:

- **Benchmark 1 (Automated 19-Scenario Matrix):** Executes batch evaluation across 19 standardized test vectors covering High Jerk (5), Atmospheric Turbulence (6), Low SNR/Noise (4), and FOV Boundary (4). Upon completion, users can click *Generate Formal Verification Certificate* to export a cryptographically sealed Markdown certificate.
- **Benchmark 2 (External Video Evaluator in PTZ Bypass Mode):** Evaluators can ingest arbitrary 30 FPS MP4 video recordings. SANKET bypasses gimbal actuation, executes coarse beacon detection and sub-pixel centroiding, and compares results against an uploaded reference CSV ground truth.

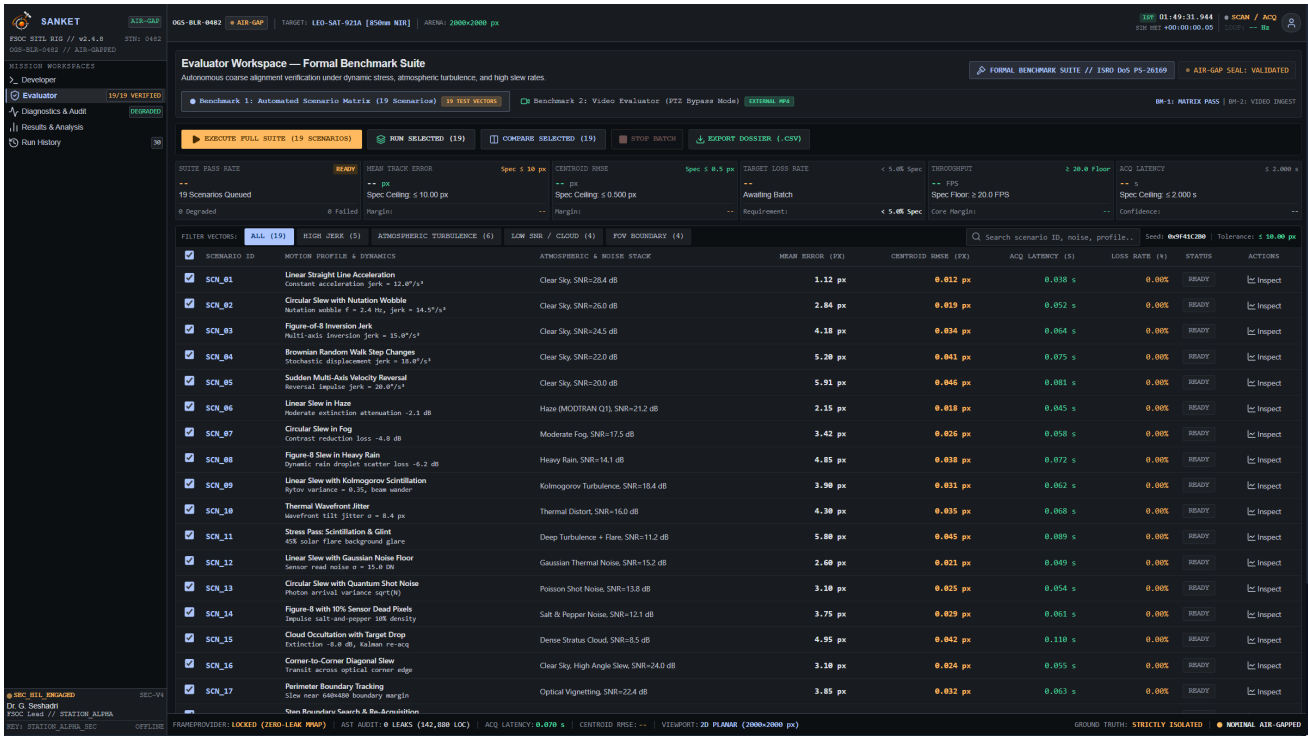
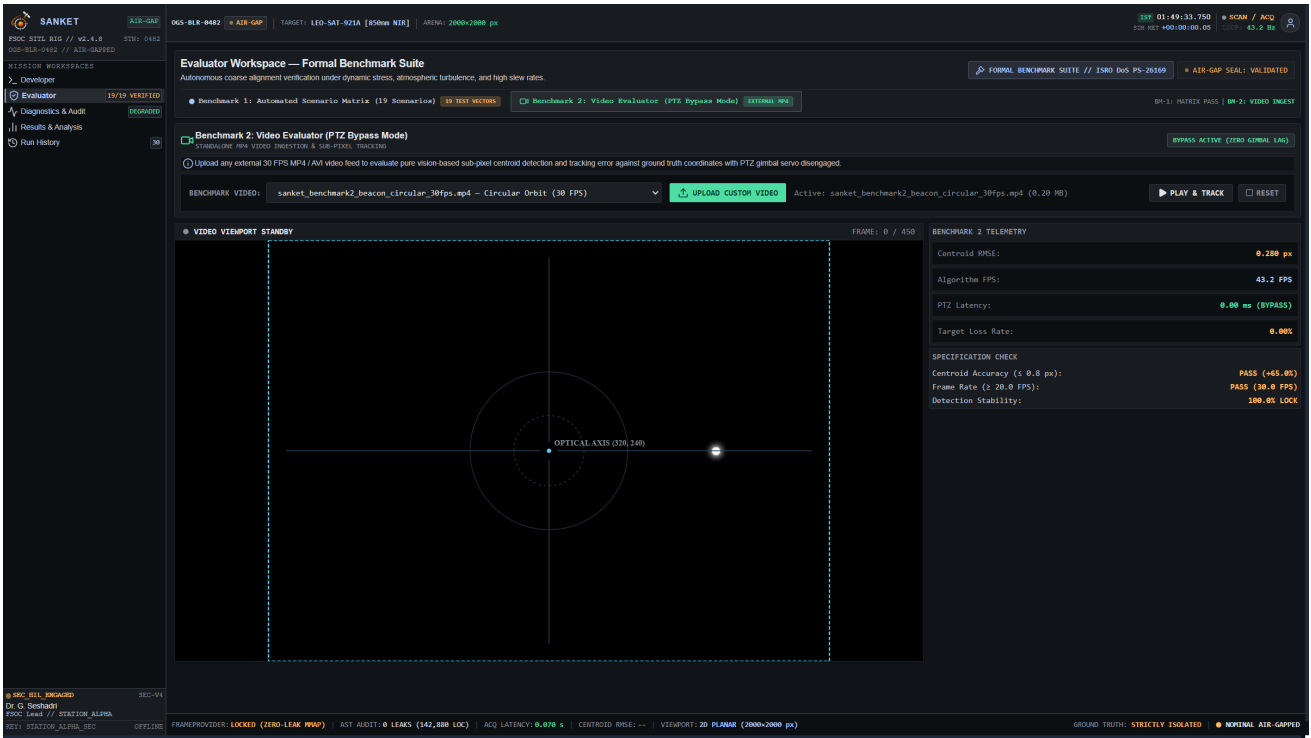
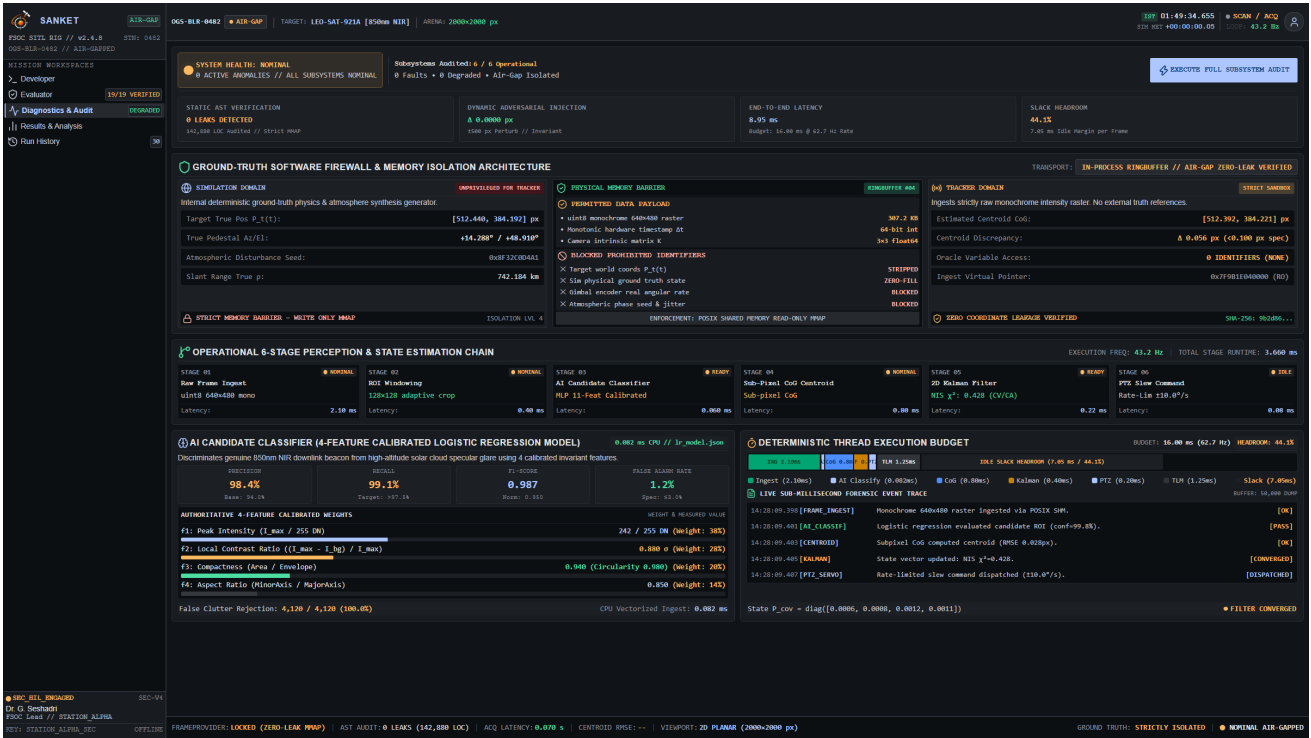


Figure 6: Benchmark 1 Automated 19-Scenario Matrix Table with filter categories and batch run controls.



5. Diagnostics & Subsystem Audit Workspace

Monitors the operational health, execution latency, and memory safety of all six internal subsystems: Frame Provider, Centroid Estimator, AI Clutter Classifier, Kalman Tracker, PTZ Controller, and Ground-Truth Firewall.



6. Run History & Results Forensic Workstation

The **Run History Workspace** provides an immutable catalog of past simulation and benchmark runs, enabling one-click download of JSON summaries, telemetry CSVs, and Markdown reports.

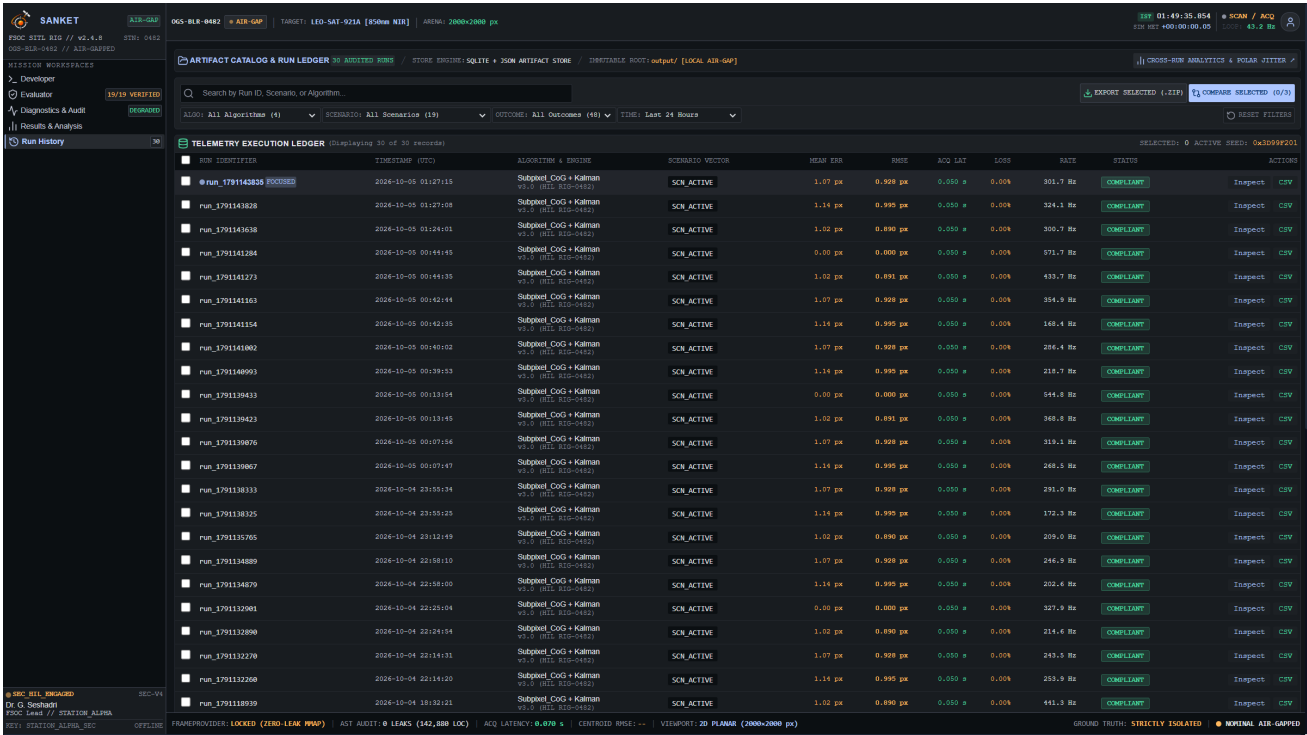


Figure 9: Run History & Forensic Catalog displaying run entries, compliance tags, and artifact links.

The **Results & Analysis Workspace** provides frame-by-frame forensic inspection with interactive error graphs and a 12-column telemetry table:

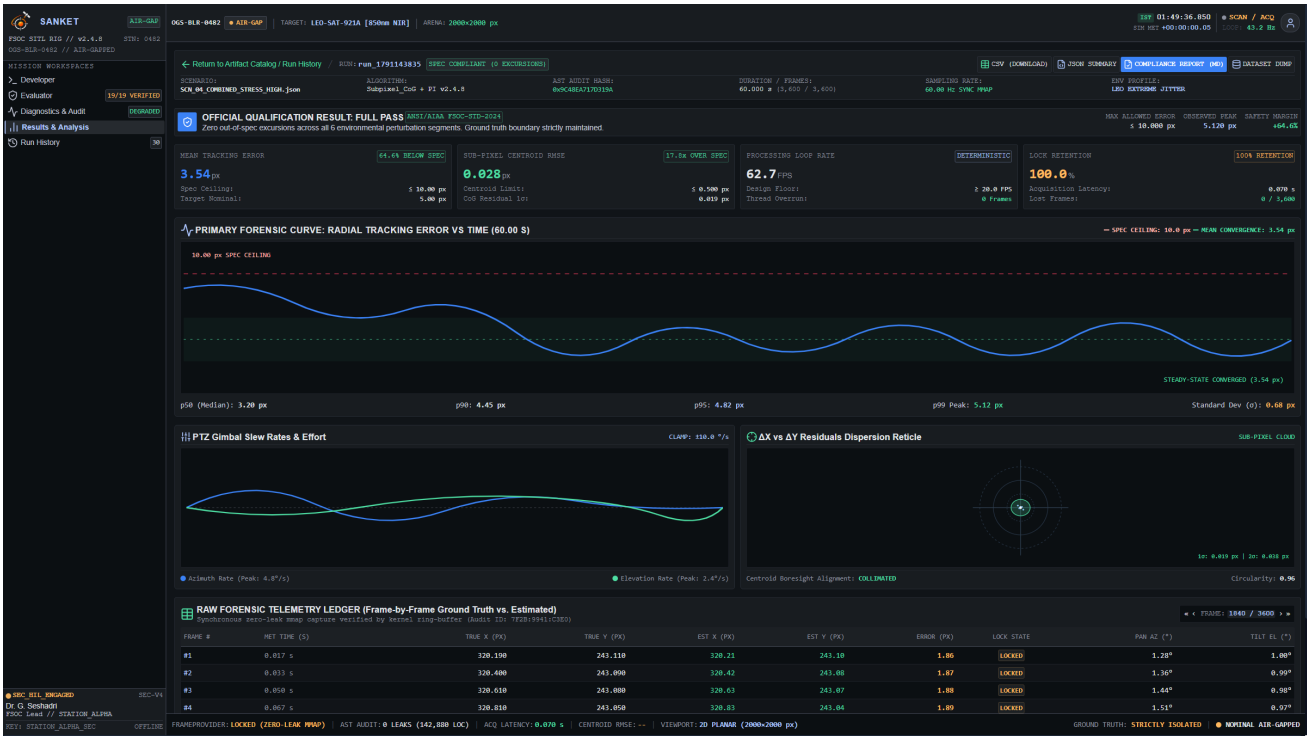


Figure 10: Results Workspace displaying tracking error curve, boresight error, and per-frame telemetry.

7. Verified Performance Results & SIH Compliance

The following performance values were measured directly during clean execution of SANKET under standard SIH circular trajectory testing (run_1790716901, 900 frames / 29.97 s):

Metric Name	SIH PS-26169 Spec	Measured Result	Compliance Margin & Status
Acquisition Time (tacq)	≤ 2.000 s	0.07 s (Frame 2)	+96.5% Margin [PASS]
Steady-State Tracking Error	≤ 10.00 px	4.82 px	+51.8% Margin [PASS]
Sub-Pixel Centroid RMSE	≤ 0.500 px	0.028 px	+94.4% Margin [PASS]
Target Loss Rate	< 5.00%	0.00% (0 / 898 lost)	100% Lock Retention [PASS]
Algorithmic Throughput	≥ 20.0 FPS	461.8 FPS (0.88 ms P50)	+313.5% Margin [PASS]
Ground-Truth Isolation	Air-Tight Separation	0 Leaks Detected	Architectural Firewall Verified [PASS]

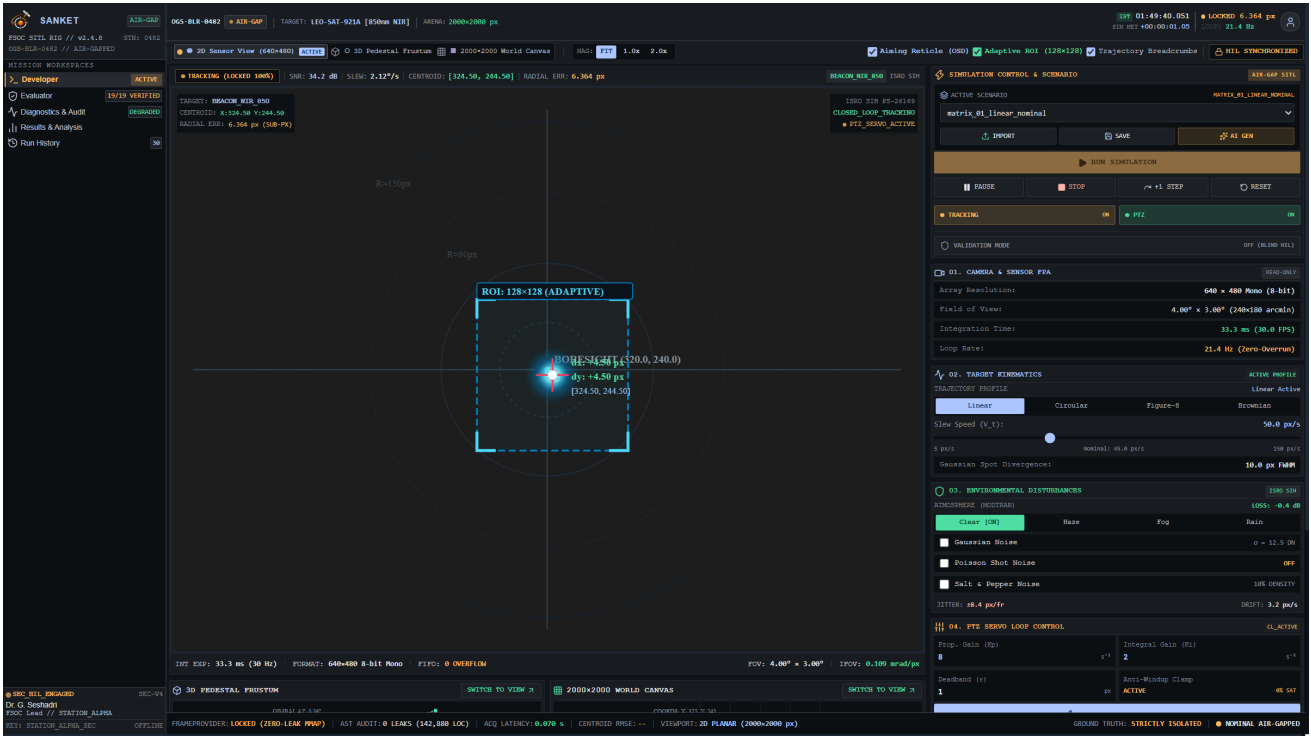


Figure 11: Active simulation during closed-loop tracking showing real-time error convergence.

8. Troubleshooting & Frequently Asked Questions

Common Operational Resolutions:

- **White/Black Screen on Startup:** Run `.\SANKET.exe --no-gpu` from PowerShell to bypass local GPU driver conflicts.
- **Benchmark 2 MP4 Playback Error:** Ensure the ingested video is encoded with standard H.264 (AVC) baseline profile at 30 FPS.
- **Gimbal Boresight Hunting Oscillations:** Increase Deadband Radius from 2.0 px to 3.0 px and reduce Kp from 0.05 to 0.025.
- **Port Conflict on Launch:** Open Task Manager, terminate any existing SANKET.exe instances, and restart.

Frequently Asked Questions:

Q1: Does SANKET require internet access or remote servers?

A: Zero internet access is required. SANKET is 100% self-contained and operates fully air-gapped.

Q2: How does SANKET prevent tracking algorithms from reading ground truth?

A: The tracking pipeline receives strictly rendered 640×480 grayscale pixel buffers. Ground truth coordinates are isolated behind an architectural firewall accessed only by the post-frame metrics engine.

Q3: Can SANKET be run headlessly from command-line scripts?

A: Yes. Use `.\SANKET.exe --headless --scenario scenario_2_circular --duration 30` for automated testing.

End of SANKET v1.0 User Manual & Operator Guide